

# Programming Fundamentals - WEEK 1

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## JOB SHEET 1 - Programming Language (Java Development Kit/JDK Installation)

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### 2.1 Experiment 1: Installing the Java Development Kit/JDK

#### Question:

1. Explain the purpose of adding the location of Java's bin folder to the PATH variable!
2. Explain the purpose of the javac command when entered in the command prompt!

#### Answer:

##### 1. The purpose of adding the location of Java's bin folder to the PATH variable

Adding the Java bin folder to the PATH variable allows Windows to find Java commands and programs more easily. It acts as a link between the operating system and the Java tools, so commands such as java and javac can be used directly in the Command Prompt without typing the full Java folder location.

##### 2. The purpose of the javac command when entered in the command prompt

The purpose of the javac command is to compile (translate) Java source code written in a `.java file`. It checks the code for errors and converts it into a `.class` file so that it can be executed by the Java Virtual Machine (JVM).

## Documentation:

```
C:\WINDOWS\system32\cmd. X + v
-profile <profile>
    Check that API used is available in the specified profile.
    This option is deprecated and may be removed in a future release.
--release <release>
    Compile for the specified Java SE release.
    Supported releases:
        8, 9, 10, ..., 24, 25, 26
-s <directory>
    Specify where to place generated source files
--source <release>, -source <release>
    Provide source compatibility with the specified Java SE release.
    Supported releases:
        8, 9, 10, ..., 24, 25, 26
--source-path <path>, -sourcepath <path>
    Specify where to find input source files
--system <jdk>|none
    Override location of system modules
--target <release>, -target <release>
    Generate class files suitable for the specified Java SE release.
    Supported releases:
        8, 9, 10, ..., 24, 25, 26
--upgrade-module-path <path>
    Override location of upgradeable modules
-verbose
    Output messages about what the compiler is doing
--version, -version
    Version information
-Werror
    Terminate compilation if any warnings occur
-Warning:<key>(<key>)*
    Specify lint categories for which warnings should terminate compilation,
    separated by comma.
    Precede a key by '-' to exclude the specified category.
    Use --help-lint to see the supported keys.

C:\Users\HP>javac -version
javac 26.0.2.1

C:\Users\HP>
```

## 2.2 Experiment 2 : Basic Java Structure

### Question:

1. Explain the function of the command `javac MyFirstJava00.java` in the experiment above!
2. Explain the function of the command `java MyFirstJava00` in the experiment above!

### Answer:

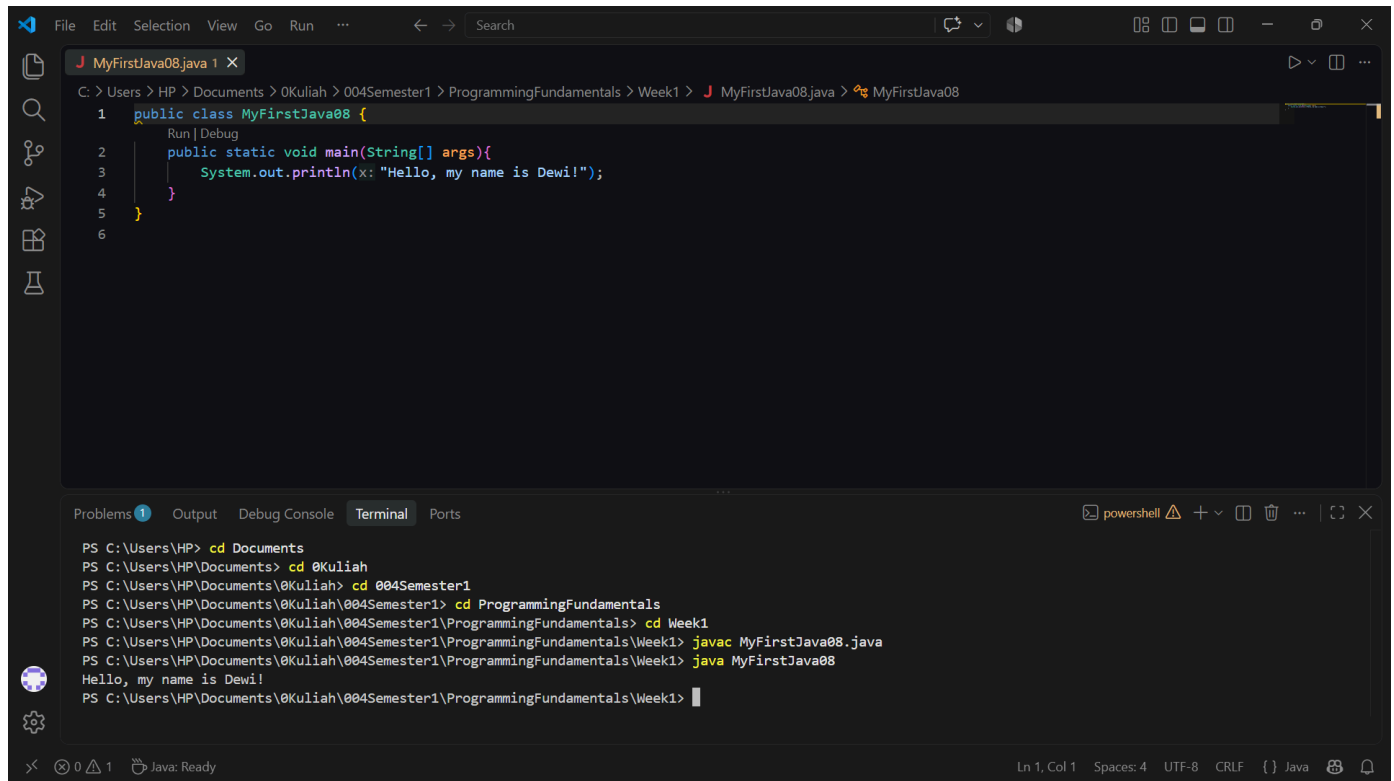
#### 1. The function of the command `javac MyFirstJava00.java` in the experiment

The `javac MyFirstJava00.java` command is used to compile the Java source code in the `MyFirstJava00.java` file. It checks the code for errors and, if there are no errors, converts the source code into a `MyFirstJava00.class` file that can be executed by the Java Virtual Machine(JVM).

#### 2. The function of the command `java MyFirstJava00` in the experiment

The `java MyFirstJava00` command is used to run the `Java` program that has already been compiled. The `java` command tells the JVM to read and execute the `MyFirstJava00.class` file. Therefore, the program can run and display the output in the terminal if there is an output command such as `System.out.println()`.

## Documentation:



The screenshot shows the Visual Studio Code editor with a Java file named `MyFirstJava08.java` open. The code is as follows:

```
1 public class MyFirstJava08 {  
2     public static void main(String[] args){  
3         System.out.println(x: "Hello, my name is Dewi!");  
4     }  
5 }  
6
```

The terminal at the bottom shows the execution of the program:

```
PS C:\Users\HP> cd Documents  
PS C:\Users\HP\Documents> cd 0Kuliah  
PS C:\Users\HP\Documents\0Kuliah> cd 004Semester1  
PS C:\Users\HP\Documents\0Kuliah\004Semester1> cd ProgrammingFundamentals  
PS C:\Users\HP\Documents\0Kuliah\004Semester1\ProgrammingFundamentals> cd Week1  
PS C:\Users\HP\Documents\0Kuliah\004Semester1\ProgrammingFundamentals\Week1> javac MyFirstJava08.java  
PS C:\Users\HP\Documents\0Kuliah\004Semester1\ProgrammingFundamentals\Week1> java MyFirstJava08  
Hello, my name is Dewi!  
PS C:\Users\HP\Documents\0Kuliah\004Semester1\ProgrammingFundamentals\Week1>
```

## 2.3 Experiment 3 : Adding an Extension in Visual Studio Code

### Question:

1. Change the `System.out.println` command to `System.out.print` on lines 6 and 7, then run the program!
2. What is the difference between `System.out.println("...");` and `System.out.print("...");`? Explain!
3. Change the commands on lines 6 and 7 to the following :

```
System.out.print("\nLearning Java is easy!");  
System.out.print("\nThe more often you practise, the better you get.");
```

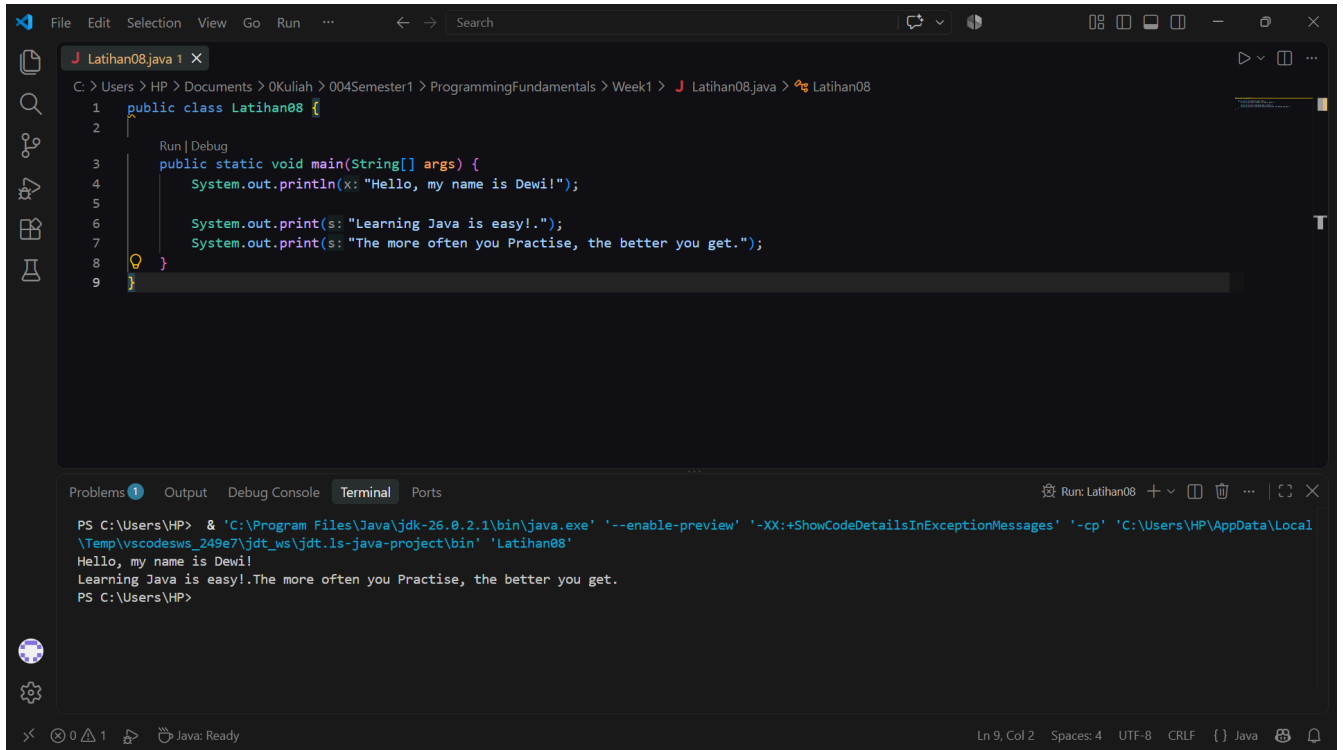
4. Explain the function of the `\n` command on lines 6 and 7!
5. On line 7, add the characters `//` so that the line becomes

```
`//System.out.print("\nThe more often you practise, the better you get.");`
```

6. Run the program and observe what happens! Explain the function of the `//` characters

### Answer:

## 7. Change the System.out.println command to System.out.print



The screenshot shows a VS Code editor with a Java file named `Latihan08.java`. The code is as follows:

```
1 public class Latihan08 {
2
3     public static void main(String[] args) {
4         System.out.println(x: "Hello, my name is Dewi!");
5
6         System.out.print(s: "Learning Java is easy!");
7         System.out.print(s: "The more often you Practise, the better you get.");
8     }
9 }
```

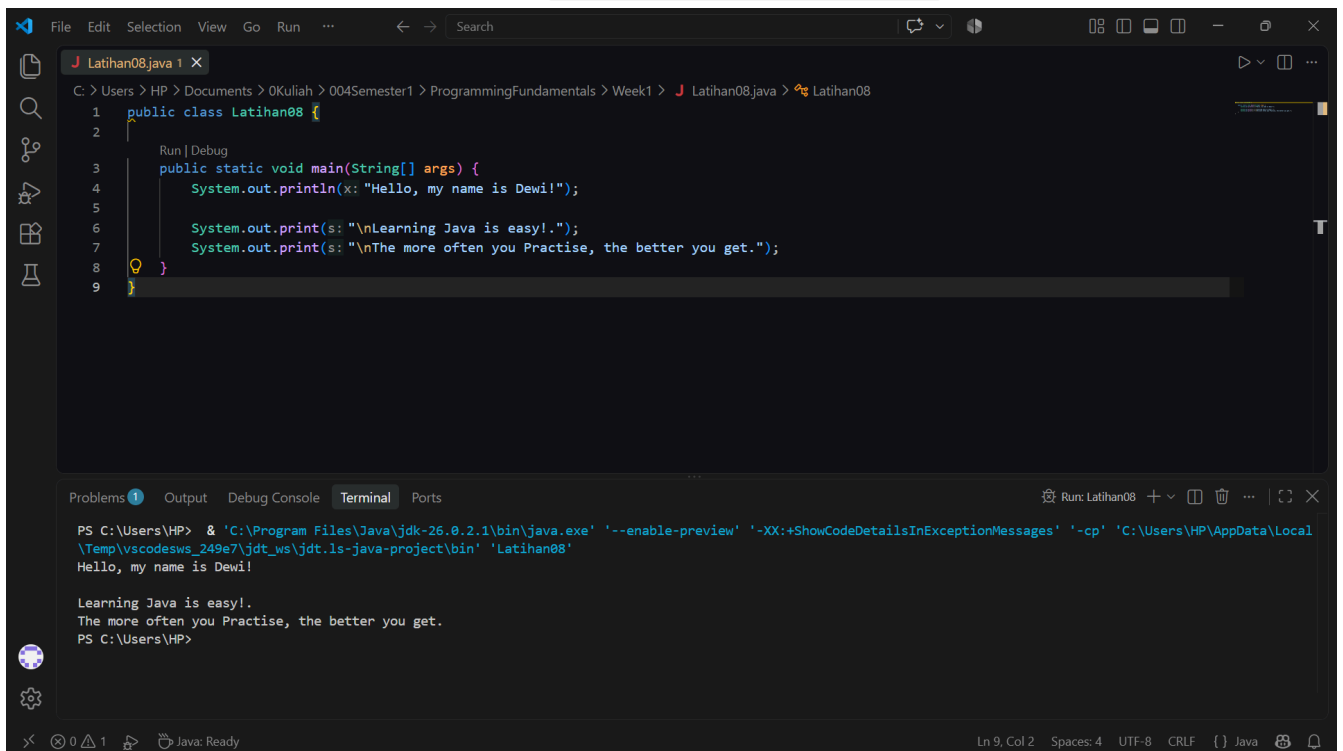
The output in the terminal is:

```
PS C:\Users\HP> & 'C:\Program Files\Java\jdk-26.0.2.1\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP\AppData\Local\Temp\vscodesws_249e7\jdt_ws\jdt.ls-java-project\bin' 'Latihan08'
Hello, my name is Dewi!
Learning Java is easy!.The more often you Practise, the better you get.
PS C:\Users\HP>
```

## 8. What is the difference between `System.out.println("...");` and `System.out.print("...");`

The difference between `System.out.println()` and `System.out.print()` is that `println()` moves to a new line after displaying the text, while `print()` keeps the output on the same line.

## 9. Change the commands on lines 6 and 7 `System.out.print("\n...");`



The screenshot shows the same VS Code editor with the modified Java code:

```
1 public class Latihan08 {
2
3     public static void main(String[] args) {
4         System.out.println(x: "Hello, my name is Dewi!");
5
6         System.out.print(s: "\nLearning Java is easy!");
7         System.out.print(s: "\nThe more often you Practise, the better you get.");
8     }
9 }
```

The output in the terminal is:

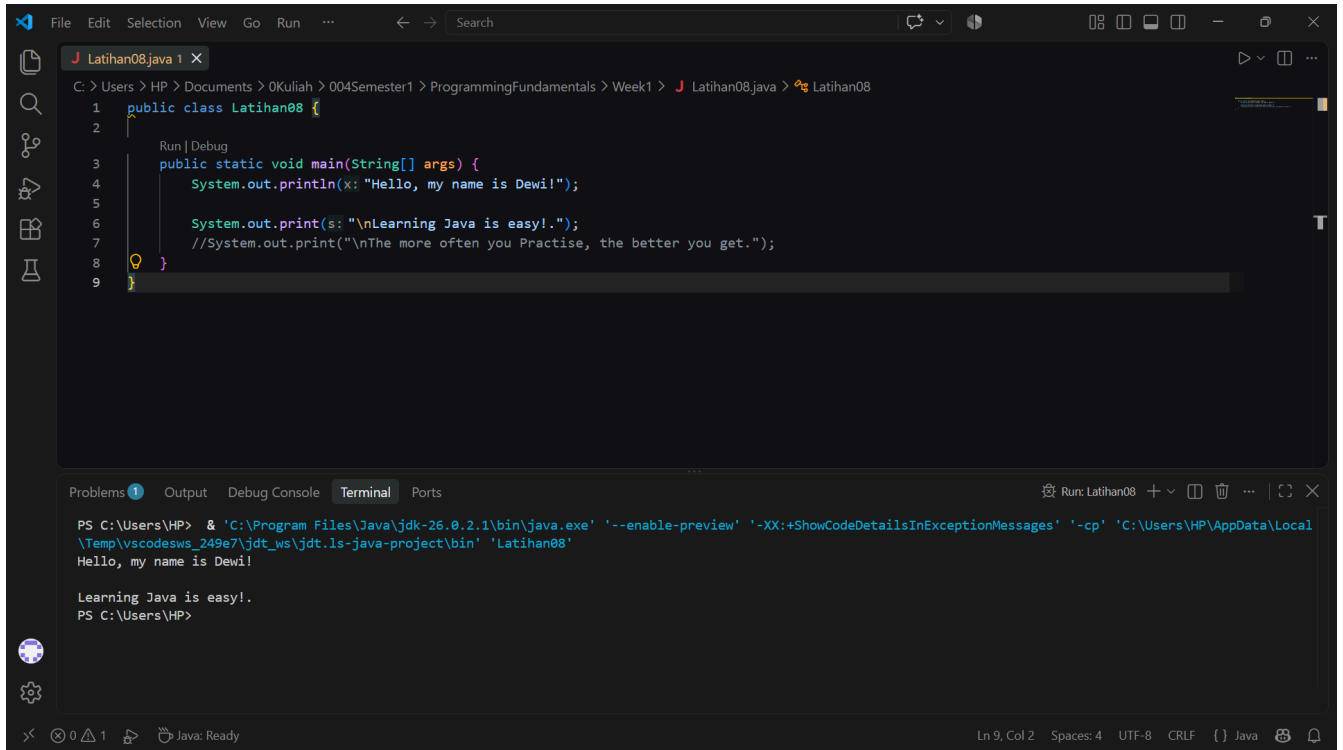
```
PS C:\Users\HP> & 'C:\Program Files\Java\jdk-26.0.2.1\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP\AppData\Local\Temp\vscodesws_249e7\jdt_ws\jdt.ls-java-project\bin' 'Latihan08'
Hello, my name is Dewi!

Learning Java is easy!.
The more often you Practise, the better you get.
PS C:\Users\HP>
```

## 10. Explain the function of the `\n` command on lines 6 and 7!

The `\n` command is used to create a new line in the output. When `\n` is used in `System.out.print(...)`, the text after `\n` will be displayed on the next line.

## 11. added the characters `//`



The screenshot shows a code editor with a Java file named `Latihan08.java`. The code is as follows:

```
1 public class Latihan08 {  
2  
3     Run | Debug  
4     public static void main(String[] args) {  
5         System.out.println(x: "Hello, my name is Dewi!");  
6  
7         System.out.print(s: "\nLearning Java is easy!");  
8         //System.out.print("\nThe more often you Practise, the better you get.");  
9     }  
}
```

Below the code editor, the `Terminal` tab shows the output of the program:

```
PS C:\Users\HP> & 'C:\Program Files\Java\jdk-26.0.2.1\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP\AppData\Local\Temp\vscode\vs_249e7\jdt_ws\jdt.ls-java-project\bin' 'Latihan08'  
Hello, my name is Dewi!  
  
Learning Java is easy!  
PS C:\Users\HP>
```

## 6. The function of the `//` characters

The `//` characters are used to make the line a comment. When `//` is placed at the beginning of a line, the Java compiler ignores everything after `//`, so the command is not executed and its output will not appear in the program.

# Assignment

**Question:** After the 3 steps above have been executed, which of the statements below are correct (more than one may apply)?

**Answer:** (a) The two balls have swapped places  
(e) tray C is empty

**Why?** Because after the three steps were executed, the two balls successfully swapped places. Tray C is also empty because it was only used as a temporary tray during the process. Therefore, the correct statements are (a) The two balls have swapped places and (e) Tray C is empty.

**Question:** Write a Java program (for example named `RobotArm00.java`, replacing 00 with your student/roll number) that prints the contents of trays A, B, and C after EVERY step (Step 1, Step 2, Step 3), ending with a conclusion line that prints which statement is correct according to your own analysis. Example output format (the dots are to be filled in based on your own analysis, not a ready-made answer):

```
Mamluatuls-MacBook-Air:~ mamluatulhaniah$ java RobotArm00  
After Step 1: Tray A = ..., Tray B = ..., Tray C = ...  
After Step 2: Tray A = ..., Tray B = ..., Tray C = ...
```

After Step 3: Tray A = ..., Tray B = ..., Tray C = ...

Conclusion: The correct statement is "..."

Mamluatuls-MacBook-Air:~ mamluatulhaniah\$

## Answer:

```
Problems 1 Output Debug Console Terminal Ports
PS C:\Users\HP>
PS C:\Users\HP> & 'C:\Program Files\Java\jdk-26.0.2.1\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP\AppData\Local\Temp\vscodesws_be2c1\jdt_ws\jdt.ls-java-project\bin' 'RobotArm08'
After Step 1: Tray A = Empty, Tray B = Moon ball, Tray C = Star ball
After Step 2: Tray A = Moon ball, Tray B = Empty, Tray C = Star ball
After Step 3: Tray A = Moon ball, Tray B = Star ball, Tray C = Empty
Conclusion: The correct statements are After the three steps were executed, the two balls successfully swapped places. Tray C is also empty because it was only used as a temporary tray during the process. Therefore, the correct statements are (a) The two balls have swapped places and (e) Tray C is empty.
PS C:\Users\HP> 
```

## How?

```
public class RobotArm08 {

    public static void main(String[] args) {
        System.out.println("After Step 1: Tray A = Empty, Tray B = Moon ball, Tray C = Star ball");
        System.out.println("After Step 2: Tray A = Moon ball, Tray B = Empty, Tray C = Star ball");
        System.out.println("After Step 3: Tray A = Moon ball, Tray B = Star ball, Tray C = Empty");
        System.out.println("Conclusion: The correct statements are After the three steps were executed, the two balls successfully swapped places. Tray C is also empty because it was only used as a temporary tray during the process. Therefore, the correct statements are (a) The two balls have swapped places and (e) Tray C is empty.");
    }
}
```